SEMESTER -VII

Course Code	Course Name	L	T	P	C
23CS7E02	PROFESSIONAL ELECTIVE- V 1. BLOCKCHAIN TECHNOLOGIES	3	0	0	3

Course Objectives:

By the end of the course, students will be able to

- Understand how blockchain systems (mainly Bit coin and Ethereum) work and to securely interact with them,
- Design, build, and deploy smart contracts and distributed applications
- Integrate ideas from block chain technology into their own projects.

Course Outcomes:

At the end of the course, the students will be able to:

CO-1: Understand evolution of Blockchain and how blockchain systems work and to securely interact with them (mainly Bit coin and Ethereum)

CO-2: Demonstrate the foundation of the Blockchain technology and understand the processes in payment and funding.

CO-3: Identify the Architecting Blockchain solutions in building Block chain applications.

CO-4: Integrate ideas on implementation of Ethereum block chain

CO-5: Choose the present landscape of Hyperledger Blockchain implementations and Understand Crypto currency markets

UNIT I

Introduction: Scenarios, Challenges Articulated, Blockchain, Blockchain Characteristics, Opportunities Using Blockchain, History of Blockchain.

Evolution of Blockchain: Evolution of Computer Applications, Centralized Applications, Decentralized Applications, Stages in Blockchain Evolution, Consortia, Forks, Public Blockchain Environments, Type of Players in Blockchain Ecosystem, Players in Market.

UNIT II

Blockchain Concepts: Introduction, Chaining of Blocks, Hashing, Merkle-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, security on blockchain, data storage on blockchain, wallets, coding on blockchain: smart contracts, peer-to-peer network, types of blockchain nodes, risk associated with blockchain solutions, life cycle of blockchain transaction.

UNIT III

Architecting Blockchain solutions: Introduction, Obstacles for Use of Blockchain, Blockchain Relevance Evaluation Framework, Blockchain Solutions Reference Architecture, Types of Blockchain Applications. Cryptographic Tokens, Typical Solution Architecture for Enterprise Use Cases, Types of Blockchain Solutions, Architecture Considerations, Architecture with Blockchain Platforms, Approach for Designing Blockchain Applications.

UNIT IV

Ethereum Blockchain Implementation: Introduction, Tuna Fish Tracking Use Case, Ethereum Ecosystem, Ethereum Development, Ethereum Tool Stack, Ethereum Virtual Machine, Smart Contract Programming, Integrated Development Environment, Truffle Framework, Ganache, Unit Testing, Ethereum Accounts, MyEtherWallet, Ethereum Networks/Environments, Infura, Etherscan, Ethereum Clients, Decentralized Application, Metamask, Tuna Fish Use Case Implementation, OpenZeppelin Contracts

UNIT V

Hyperledger Blockchain Implementation, Introduction, Use Case – Car Ownership Tracking, Hyperledger Fabric, Hyperledger Fabric Transaction Flow, FabCar Use Case Implementation, Invoking Chain code Functions Using Client Application. Advanced Concepts in Blockchain: Introduction, Inter Planetary File System (IPFS), Zero-Knowledge Proofs, Oracles, Self-Sovereign Identity, Blockchain with IoT and AI/ML Quantum Computing and Blockchain, Initial Coin Offering, Blockchain Cloud Offerings, Blockchain and its Future Potential.

Text Books:

1) Ambadas, ArshadSarfarzAriff, Sham "Blockchain for Enterprise Application Developers", Wiley

2) Andreas M. Antonpoulos, "Mastering Bitcoin: Programming the Open Blockchain", O'Reilly.

Reference Books:

1) Blockchain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Bambara, Paul R. Allen, McGraw Hill.

2) Blockchain: Blueprint for a New Economy, Melanie Swan, O'Reilly.

E-Resources:

https://github.com/blockchainedindia/resources

Faculty coordinators

1. P. Neela Sundari

2. T. Soundarya

HoD - CSE

Prof. K. Kamesh

Professor & H.O.D

Dept. of Computer Science & Engg.

AR & KSR Institute of Technoloy & Science

Venkatampadu GUNTUR-17